

# Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects\*

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## Abstract

To estimate the dynamic effects of an absorbing treatment, researchers often use two-way fixed effects regressions that include leads and lags of the treatment. We show that in settings with variation in treatment timing across units, the coefficient on a given lead or lag can be contaminated by effects from other periods, and apparent pretrends can arise solely from treatment effects heterogeneity. We propose an alternative estimator that is free of contamination, and illustrate the relative shortcomings of two-way fixed effects regressions with leads and lags through an empirical application.

*Keywords:* difference-in-differences, two-way fixed effects, pretrend test

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following differences. First, in the setting where there is no never-treated group, our method uses the last cohort to be treated as a control group, whereas Callaway and Sant’Anna (2020a) use the set of not-yet-treated cohorts. Our method and theirs thus rely on different, but non-nested parallel trends assumptions. Second, our estimation method can be cast as a regression specification and thus may be more familiar to applied researchers. However, a third difference is that the procedure of Callaway and Sant’Anna (2020a) allows for conditioning on time-varying covariates. de Chaisemartin and D’Haultfœuille (2020) and Goodman-Bacon (2018) respectively propose alternative estimators and diagnostic tools for estimation of causal effects in staggered settings, but do not consider the estimation of the dynamic path of treatment effects as we do.

## 2 Event studies design

In this section we first formalize the “event studies design”. As discussed in Section 2.3, based on how this term is deployed in the empirical literature, an *event study* design is a *staggered adoption* design where units are treated at different times, and there may or may not be never treated units. It also nests a *difference-in-differences* design, where units are either first treated at time  $t_0$  or never treated.

Specifically, we consider a setting with a random sample of  $N$  units observed over  $T + 1$  time periods, where  $T$  is fixed. For each  $i \in \{0, \dots, N\}$  and  $t \in \{0, \dots, T\}$ , we observe the outcome  $Y_{i,t}$  and treatment status  $D_{i,t} \in \{0, 1\}$ :  $D_{i,t} = 1$  if  $i$  is treated in period  $t$  and  $D_{i,t} = 0$  if  $i$  is not treated in period  $t$ . Throughout we assume that the observations  $\{Y_{i,t}, D_{i,t}\}_{t=0}^T$  are independent and identically distributed (i.i.d.).

In the general case of event studies we focus on an *absorbing* treatment such that the treatment status over time is a non-decreasing sequence of zeros and then ones, i.e.  $D_{i,s} \leq D_{i,t}$  for  $s < t$ . We can thus uniquely characterize a treatment path by the time period of the initial treatment, denoted with  $E_i = \min \{t : D_{i,t} = 1\}$ . If unit  $i$  is never treated i.e.  $D_{i,t} = 0$  for all  $t$ , we set  $E_i = \infty$ . Based on when they first receive the treatment, we can also uniquely categorize units into disjoint cohorts  $e$  for  $e \in \{0, \dots, T, \infty\}$ , where units in cohort  $e$  are first treated at the same time  $\{i : E_i = e\}$ .

We define  $Y_{i,t}^e$  to be the potential outcome in period  $t$  when unit  $i$  is first treated in time period  $e$ . We define  $Y_{i,t}^\infty$  to be the potential outcome if unit  $i$  never receives the treatment, which we call the “baseline outcome”. Since the timing of the initial treatment uniquely characterizes one’s treatment path, we can

treatment is

$$CATT_{e,\ell} = E[Y_{i,e+\ell} - Y_{i,e+\ell}^{\infty} \mid E_i = e]. \quad (3)$$

Each  $CATT_{e,\ell}$  represents the average treatment effect  $\ell$  periods from the initial treatment for the cohort of units first treated at time  $e$ . We shift from calendar time index  $t$  to relative period index  $\ell$  which denotes the periods since treatment; for cohort  $e$ ,  $\ell$  ranges from  $-e$  to  $T - e$  because we observe at most  $e$  periods before the initial treatment and  $T - e$  periods after the initial treatment. Relative periods allow us to compare cohorts while holding their exposure to the treatment constant.

## 2.2 Identifying assumptions

With the above definitions, we formalize three potential identifying assumptions for outcomes of interest in our event study design. The first assumption is a generalized form of a parallel trends assumption. The second assumption requires no anticipation of the treatment. The third assumption imposes no variation across cohorts. For each assumption, we first discuss its meaning and then compare it with similar assumptions made in the literature interpreting two-way fixed effects regressions. Later in Section 3 we interpret the relative period coefficients  $\mu_{\ell}$  from two-way fixed effects regressions under different combinations of these assumptions.

**Assumption 1.** (Parallel trends in baseline outcomes.) *For all  $s \neq t$ , the  $E[Y_{i,t}^{\infty} - Y_{i,s}^{\infty} \mid E_i = e]$  is the same for all  $e \in \text{supp}(E_i)$ .*

If an application includes never-treated units so that  $\infty \in \text{supp}(E_i)$ , we need to especially consider whether these never-treated units satisfy the parallel trends assumption. Never-treated units are likely to differ from ever-treated units in many ways, and may not share the same evolution of baseline outcomes. If the never-treated units are unlikely to satisfy the parallel trends assumption, then we should exclude them from the estimation to avoid violation of this assumption.

While common in the applied literature, the parallel trends assumption is strong and oftentimes violated. For example, Ashenfelter (1978) documented that participants in job training programs experience a decline in earnings prior to the training period (Ashenfelter's dip). The timing of job training is dependent on the evolution of individual's baseline earnings, and this scenario therefore does not satisfy the parallel trends assumption. Proposition 1 is the only result in this paper derived without this assumption, but there is an

at least  $l$  periods relative to hospitalization by its sample analog. Third, we form IW estimates  $\widehat{v}_\ell$  by taking weighted averages of  $\widehat{\delta}_{e,\ell}$  (returned from step one) with sample cohort share (returned from step two) as weights.

In Table 3, we report the FE estimates  $\widehat{\mu}_\ell$  and the IW estimates  $\widehat{v}_\ell$ , as well as the underlying  $CATT_{e,\ell}$  estimates  $\widehat{\delta}_{e,\ell}$ . In this application, the FE estimates and IW estimates happen to be very similar in their magnitude. The conclusion from Dobkin et al. (2018) based on FE estimates similar to ours still holds: we find a substantial and persistent decline in the earnings due to hospitalization, and the increase in out-of-pocket spending is transitory and small in comparison.

However, the FE estimates still suffer from non-convex and non-zero weighting as we show in Section 5.3, which can lead to interpretability issues depending on the amount of treatment effect heterogeneity. Even though for both outcomes  $\widehat{\mu}_0$  falls in the convex hull of its underlying  $CATT_{e,0}$  estimates,  $\widehat{\mu}_{-2}$  turns out to be outside the convex hull of its underlying  $CATT_{e,-2}$  estimates. In contrast, by construction, the IW estimates  $\widehat{v}_\ell$  fall within the convex hull of its underlying  $CATT_{e,\ell}$  estimates and are unaffected by  $CATT_{e,\ell'}$  estimates from other periods  $\ell' \neq \ell$ . Thus, they have an interpretation as an average effect of the treatment on the treated  $\ell$  periods after initial treatment.

## 6 Conclusions

This paper analyzes the behavior of relative period coefficients  $\mu_\ell$  on the indicator for being  $\ell$  periods away from the treatment from two-way fixed effects regressions in settings with variation in treatment timing and treatment effects heterogeneity. For dynamic treatment effects, researchers are usually interested in estimating some average of treatment effects from  $\ell$  periods relative to the treatment, and it is common to report the coefficient estimate  $\widehat{\mu}_\ell$  assuming that interpretation is valid. However, we show that in the presence of heterogenous treatment effects, the coefficient  $\mu_\ell$  does not necessarily capture the dynamic treatment effect as it can fall outside the convex hull of  $CATT_{e,\ell}$  from its corresponding period  $\ell$  and may pick up spurious terms consisting of treatment effects from periods other than  $\ell$ . This negative result is based on the decomposition of the coefficients  $\mu_\ell$  as a linear combination of cohort-specific average treatment effects on the treated  $CATT_{e,\ell}$ . The weights in the linear combination can be non-convex, and non-zero on relative periods  $\ell' \neq \ell$ .

Given these negative results on two-way fixed effects regression estimators, we propose “interaction-

weighted” (IW) estimators for estimating dynamic treatment effects. The IW estimators are formed by first estimating  $CATT_{e,\ell}$  with a regression saturated in cohort and relative period indicators, and then averaging estimates of  $CATT_{e,\ell}$  across  $e$  at a given  $\ell$ . These  $CATT_{e,\ell}$  are identified under parallel trends and no anticipation assumptions. These estimators are easy to implement and robust to heterogeneous treatment effects across cohorts; the IW estimator associated with relative period  $\ell$  is guaranteed to estimate a convex average of  $CATT_{e,\ell}$  using weights that are sample share of each cohort  $e$ .

Finally, we illustrate the empirical relevance of our results by estimating the dynamic effect of hospitalization on the out-of-pocket medical spending and labor earnings using a setup similar to Dobkin et al. (2018). We find non-convex and non-zero weights for two-way fixed effects regression in this example, and show that the resulting estimates indeed sometimes fall outside the convex hull of the underlying  $CATT$  estimates due to contamination by treatment effects from other relative periods. IW estimates, on the other hand, are weighted averages of the underlying  $CATT$  estimates with weights representative of cohort share.

More broadly, our paper suggests that researchers can do more in this context to assess the validity of underlying assumptions and how violations may impact their estimates. We demonstrate the sensitivity of two-way fixed effects regressions to underlying assumptions and provide researchers with tools to assess and address these issues. Specifically, for average treatment effects estimated using two-way fixed effects regressions we recommend that empirical researchers directly estimate the underlying weights on cohort-specific average treatment effects using our proposed auxiliary regressions. This exercise allows empirical researchers to assess the degree of potential contamination under any assumed structure of treatment effects heterogeneity. We also recommend empirical researchers consider estimation methods more robust to treatment effects heterogeneity such as the IW estimator we propose. Developing additional tools for applied researchers to identify and estimate interpretable parameters of interest is a promising avenue for future research.